MODULE 4

Access Control Lists

Network Defense | Cisco Networking Academy

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Topics:

- What is an ACL?
- Packet Filtering
- Numbered and Named ACLs
- How an ACL Operates
- Summary
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What Is an ACL?

- A list of rules a router uses to allow or block network traffic
- Checks every packet before it enters or leaves a network
- Works like a checklist — compares traffic to each rule, in order
- Can filter traffic by IP address, protocol, or port number

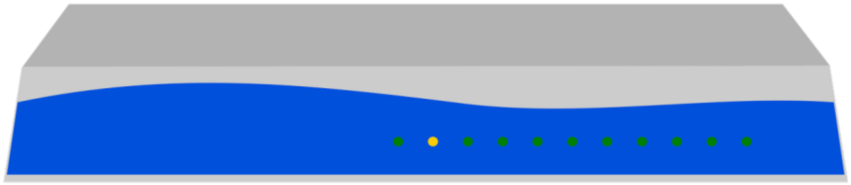
Think of an ACL like a security guard checking a list before letting anyone through the door.

Example:

A router with an ACL can allow traffic only from 192.168.1.0/24, and block everyone else.

Packet Filtering

- ACLs filter packets by looking at the source/destination address, protocol, or port
- Every packet gets one of two outcomes: Permit (let it through) or Deny (block it)
- Filtering happens at a router interface, on traffic coming in or going out
- Helps stop unwanted or unsafe traffic before it reaches the network



A router applies its ACL rules to packets moving through its interfaces.

Example:

A company can block Telnet traffic (port 23) from outside, but still allow normal web traffic (port 80/443).

Numbered and Named ACLs

Numbered ACL

- Identified by a number, e.g. 1–99 (standard) or 100–199 (extended)
- Quick and simple to create
- Harder to remember what each list actually does

Named ACL

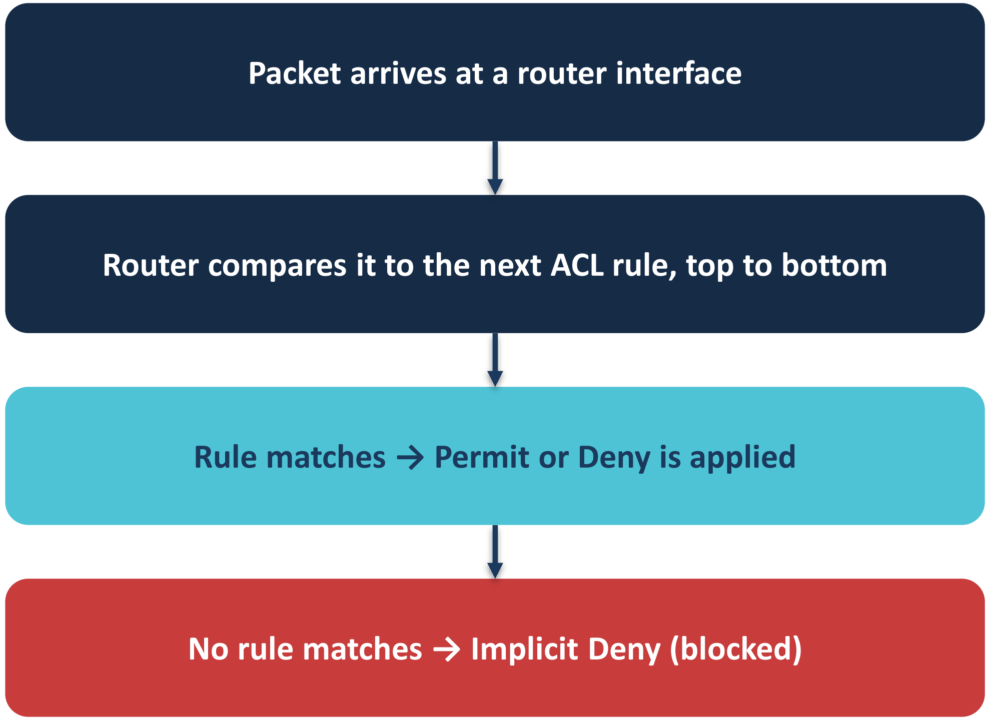
- Identified by a name, e.g. BLOCK_TELNET
- Easier to read and manage
- Individual lines can be edited without deleting the whole list

Example:

access-list 10 permit 192.168.1.0   vs.   ip access-list standard ALLOW_LAN

How an ACL Operates

- The router reads the ACL from top to bottom, rule by rule
- As soon as a packet matches a rule, that rule is applied immediately
- If no rule matches, the packet is blocked automatically — called “implicit deny”
- Rule order matters — place specific rules before general ones



Example:

If rule 1 denies all traffic and rule 2 permits one address, rule 2 is never reached — order really matters!

Summary

- An ACL is a list of rules a router uses to permit or deny traffic
- ACLs filter packets by address, protocol, or port number
- Numbered ACLs are quick to set up; named ACLs are easier to manage
- Routers check rules from top to bottom and apply the first match
- Anything that matches no rule is blocked by default (implicit deny)

References

Cisco Networking Academy

https://www.netacad.com

Cisco

https://www.cisco.com/c/en/us/support/docs/security/ios-firewall/23602-confaccesslists.html

NIST

https://www.nist.gov/cyberframework